Efficiency across generations

A reading guide for the housing model   September 5, 2026

1. Whose welfare enters the comparison?

An overlapping-generations economy contains households with finite lives, but an endless succession of households. Two periods of life do not mean that the economy contains only two generations in total.

For a full Pareto comparison, include the initial old, the initial young, and every future cohort. Compare each household's lifetime utility under the reform with its utility along the original equilibrium path, starting from the same inherited situation. For the initial old, their past is fixed, so only their remaining life can change. With heterogeneous households, an average improvement within a cohort is insufficient.

For this discussion only, let ΔU_i denote the change in the lifetime utility of person i; this does not replace the model's value-function notation. A Pareto improvement requires

$$\Delta U_i \geq 0 \quad \text{for every person in every cohort}, \qquad \Delta U_j > 0 \quad \text{for at least one person } j.$$

There is no need to sum these utilities or choose social discount weights. Those choices arise when specifying a social objective that may trade one person's loss against another's gain.

Balasko and Shell (1980, p. 286) give a precise definition for an infinite sequence of generations. They distinguish full Pareto optimality from the weaker requirement of admitting no improvement that changes only finitely many households' allocations. Their terminology for this weaker requirement is specific to that paper. The economic distinction is what matters here.

Including a cohort does not require changing its allocation. If a feasible reform leaves its entire lifetime allocation unchanged, its welfare comparison is already settled. A finite construction can therefore prove inefficiency of the full economy. But showing that no finite construction works does not exclude an improvement involving infinitely many cohorts.

2. A small example

Consider a separate teaching economy with one young and one old person at each date. Each person receives two units of goods when young and one when old. Goods cannot be stored, there are no privately traded assets, and utility is $\log c^Y + \log c^O$. Without transfers, consumption is $(2, 1)$.

Suppose every young person gives half a unit to the current old person, at every date. Each young person's lifetime consumption becomes $(1.5, 1.5)$:

$$\log 1.5 + \log 1.5 = \log 2.25 > \log 2 = \log 2 + \log 1.$$

The initial old person also gains, consuming 1.5 instead of 1. Consumption still totals three units at every date. Everyone gains over their own life, although every young person initially gives something up.

If the arrangement stops, the last young person who pays does not receive the later transfer. The absence of a last generation matters. This original example illustrates Samuelson's intergenerational-transfer intuition; it is not a specification of our housing economy.

3. Three questions in the literature

Can competition be inefficient simply because generations overlap? Samuelson (1958) develops the consumption-loan economy and the role of transfers between successive generations. Balasko and Shell (1980) make the welfare distinctions precise: in their frictionless exchange economy, competitive allocations cannot be improved by changing only finitely many households, yet need not be fully Pareto optimal. They allow competitive trade in dated goods. Thus the classical OLG issue cannot be reduced to the usual statement that an insurance market is missing.

Can an economy accumulate too much capital? Diamond (1965, sections 3–4) extends OLG analysis to production. Beyond the Golden Rule capital stock, reducing capital accumulation can release goods for consumption and permit a Pareto improvement. In the basic deterministic model, the familiar comparison between the return on capital and population growth diagnoses this possibility. Diamond also distinguishes comparing stationary allocations from finding an improvement that respects initial conditions.

Our illustrative housing model has a fixed housing stock and an externally given bond price. We have not identified a Diamond overaccumulation mechanism. Its housing-allocation question must be established from its own preferences, financial restrictions, and resource constraints. A comparison of interest and growth rates would not establish our proposed result.

How can we separate efficiency from redistribution along a transition? Auerbach and Kotlikoff (1987, chapter 5, section B.3, pp. 62–64) introduce a hypothetical *Lump Sum Redistribution Authority*. It compensates cohorts while the reform and compensation are jointly evaluated in general equilibrium. The authority balances its present-value budget, with assets or debt allowed along the path. Transfers go to existing cohorts initially and to later cohorts on entry. This is an efficiency-measurement device, not a requirement to propose a practical institution.

That is a useful precedent for our exercise. It does not establish that our particular transfers are feasible: cash before a housing purchase, future tax enforcement, and the treatment of initial owners remain model-specific assumptions to assess.

4. Full-path welfare and steady-state welfare

These comparisons answer different questions:

Comparison	What must be established
Pareto improvement	No initial or future household loses; someone gains.
Steady-state welfare	A specified household is better off at the new stationary allocation. Transitional cohorts still require evaluation.
Weighted welfare	Gains and losses are aggregated using an explicit social objective and a well-defined treatment of the infinite horizon.

For example, a reform might raise the lifetime utility of every cohort born after year 20 while harming some earlier cohorts. That is a long-run welfare gain. The uncompensated reform is not a Pareto improvement. Whether feasible compensation can remove the losses is a further question.

5. What this means for our planner

The welfare comparison and the planner's powers are separate choices. We can evaluate every generation while limiting the planner to transfers at one date. We can also evaluate every generation while allowing a committed transfer sequence. The latter gives the planner more ways to rearrange resources; it does not change the meaning of "no household loses."

Bisin's *General Equilibrium Theory* lecture notes (2014, section 3.2.4) are a useful introduction to constrained efficiency. They specify the planner's instruments and let subsequent markets clear at endogenous prices. The incomplete-asset-market literature, including Geanakoplos and Polemarchakis (1986), provides conditions under which asset redistribution can improve welfare after spot prices adjust. These results depend on their market and heterogeneity assumptions; they are not an automatic theorem for our deterministic housing OLG model.

A one-time gift is one possible restriction, not the default definition of constrained efficiency. Our local obstruction to that instrument does not settle broader transfer classes. The committed-transfer result remains conditional on its financing and ownership permissions. The literature helps frame these choices; each of our proofs still needs its own verification.

Fertility introduces a further question. Holding individual fertility and cohort sizes fixed lets us compare the same people. When fertility changes, some people exist in one allocation but not the other. Golosov, Jones, and Tertilt (2007, section 3.1) study alternative efficiency definitions for this setting. Their A-efficiency compares people alive in both allocations over the entire history; it does not mean only people alive at the reform date. Their P-efficiency includes potential people and requires specifying welfare when they are not born. A larger population alone is not a welfare ranking.

Our housing-efficiency claim and the fertility prediction can remain separate. Predicting a larger population does not itself require a welfare ranking across populations.

6. A short reading route

For our immediate question, begin with Auerbach and Kotlikoff, pp. 62–64. Use Samuelson and Balasko–Shell for foundations, Diamond for capital accumulation, and Bisin for constrained efficiency. Defer population welfare until we need it. The links below lead to primary texts or publication records.

- Samuelson (1958), "An Exact Consumption-Loan Model of Interest with or without the Social Contrivance of Money," *JPE* 66, 467–482.
- Balasko and Shell (1980), "The Overlapping-Generations Model, I: The Case of Pure Exchange without Money," *JET* 23, 281–306. Definitions: p. 286.
- Diamond (1965), "National Debt in a Neoclassical Growth Model," *AER* 55, 1126–1150. Begin with pp. 1128–1129.
- Auerbach and Kotlikoff (1987), *Dynamic Fiscal Policy*. Chapter 5, section B.3, pp. 62–64.
- Bisin (2014), *General Equilibrium Theory*, lecture notes, section 3.2.4. Technical background: Geanakoplos and Polemarchakis (1986), "Existence, Regularity, and Constrained Suboptimality of Competitive Allocations When the Asset Market Is Incomplete."
- Golosov, Jones, and Tertilt (2007), "Efficiency with Endogenous Population Growth," *Econometrica* 75, 1039–1071. Section 3.1.

7. A benchmark drawn from the literature

The author's criterion is to use established fertility and OLG methods for the authority's powers. The literature provides both a way to measure gains after compensation and examples of public transfers overcoming private financial restrictions. These are related precedents with different roles.

Compensation along the transition. Auerbach and Kotlikoff supply the reference method described on page 2. It has also been used with endogenous fertility: Okamoto's 2021 discussion paper, *Intergenerational Earnings Mobility and Demographic Dynamics*, section 3.1, compensates existing households and makes transfers to future households on entry, with aggregate present-value balance. This is a direct application, not a general theorem resolving welfare across different populations.

Public transfers can overcome private transfer restrictions. Schoonbroodt and Tertilt (2014), *Journal of Economic Theory* 150, 551–582, study this in a fertility OLG model. Sections 5.2–5.3 use fertility-dependent pensions, or fertility subsidies financed by debt and subsequent taxes, to implement an efficient allocation despite private transfer constraints. Their mechanism concerns parents and children's resources. It is not a housing theorem. They also explicitly distinguish implementing an efficient allocation from improving upon the initial allocation without losers.

Public financing across a person's life has an OLG precedent. Boldrin and Montes (2005), *Review of Economic Studies* 72, 651–664, link education finance and pensions to replace missing education credit. Young people receive financing and later repay through taxes supporting older lenders. The linked source is the Federal Reserve's working-paper abstract and publication record. Bishnu, Garg, Garg, and Ray (2023), *Journal of Economic Dynamics and Control* 153, 104697, examine an education-pension package with endogenous fertility; their government taxes the middle-aged and balances its budget each period. These papers support the economic role of public financing, with instruments particular to education.

Recommendation for our housing comparison. Use a government that can commit to dated lump-sum taxes and transfers, subject to an explicit public budget and financing constraint. Households choose housing, tenure, and saving at market prices. Preserve the physical housing restrictions, including the agreed tenure segmentation. Hold individual fertility fixed and test whether initial claimants and all household cohorts can be at least as well off, with a strict gain somewhere. This is a proposed adaptation of the established intergenerational-transfer approach.

What must remain explicit. The original compensation authority pays future households once on entry. Our construction uses cash before purchase and an additional payment or tax when old. With borrowing constraints, equal present values do not make these timings equivalent. Eligibility for transfers, eligibility of cash for down payments, and taxation or compensation of initial title owners also require specification. The existing passive outside owner is our accounting assumption, not part of the cited benchmark.

Government financing can reduce the force of a private borrowing limit while leaving the mortgage formula unchanged. A result under that permission should state it plainly. The next proof assessment should map these permissions to the construction already available; the citations themselves do not establish its constrained-inefficiency claim.